

Jigsaw-Inspired Modular Architected Materials with Tailorable Stiffness and Programmable Reconfiguration for Adaptive Flow Regulations

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The emergence of reconfigurable materials provides new avenues to design functional devices and structures at various length scales, but their post-fabricated geometries usually remain unchangeable, showing low reconfigurability and poor adaptability toward multiple functionalities. Herein, a new class of modular architected materials is introduced. Inspired by the game of jigsaw puzzle, modular building blocks are designed and fabricated. Their versatility is demonstrated by an easy-to-assemble routine to form geometric designs with desirable properties for a specific function. Guided by numerical simulations and experimental tests, the merit of the modular architected material is exemplified by designing adaptive flow regulators. By strategically arranging units and joints, the designed structures exhibit tunable stiffness and programmable reconfigurations, resulting in an adaptive flow regulation compared to their rigid counterpart. Overall, the integrated approach allows customized structural designs with less material waste, enriching the versatility of architected materials for multifunctional devices.

characterizations. This issue becomes particularly critical for the customized design of nonhomogeneous architected materials with nonperiodic, asymmetric, gradient, and hierarchical geometric features.

To overcome such limitations, we introduce a new class of modular architected materials that are assembled by repeated simple building blocks and exhibits on-demand customized structural forms. Our strategy was inspired by the well-known game of jigsaw puzzle^[33–37] that requires the assembly of large amounts of interlocking pieces to form a complete picture that can be disassembled for reuse. As a proof of concept, we utilize this strategy to design adaptive flow regulators. Our previous work^[38] has showcased a proof-of-concept flow regulator but only a few simple geometric designs using flexi-

ble joints were discussed and low attention was given to how the flow was regulated by our shape-changing structures. In this study, we exploit the design and assembly of modular architected materials and introduce a stiffness-tuning elastic joint induced by dual-material printing on structural units. Careful attention was given to the interaction between flow and regulator based on improved numerical models. Guided by experiments and simulations, we demonstrate flow-regulating units with tailorable stiffness and programmable reconfigurations through an integrated material–structural design. Our proposed modular approach also helped us accelerate the characterization of different geometric designs and joint properties featuring a direct assembly of building blocks without printing entire new specimens from scratch. While we focused on desktop-scale flow regulators, our strategy can be readily scaled to a wide range of flow control scenarios from microscale valves to large-scale weirs.^[38–42] The proposed modular design also opens new avenues for the assembly and disassembly of instability-based, auxetic, and other-type of architected materials.

1. Introduction

Origami engineering has recently become a multidisciplinary research field, creating a new playground for scientists and engineers to develop next-generation functional materials and structures.^[1–4] Applying origami principles allows material systems to switch between various configurations,^[5–13] which could enable unprecedented mechanical properties and multiple novel functionalities.^[14–32] However, since most existing reconfigurable architected materials have relied on fabrication techniques such as laser cutting or 3D printing, their resulting units are constrained to severely limit geometric forms once fabricated and suffer low flexibility to customize specific properties toward a given function. Testing a new design requires a rerun of the fabrication cycle that processes raw materials in an unsustainable manner and meanwhile lowers the efficiency of material

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2. Results and Discussions

2.1. Hybrid Unit Design and Assembly

To illustrate the proposed integrated material–structural modular approach, we begin with a square block ($h_b = 12.5$ mm) with interlocking notches ($b_n = 5$ mm, $b_t = 2.5$ mm, $h_n = 2.5$ mm) yet

a variety of basic geometries can be used alternatively (**Figure 1**). Here, two S blocks could form either a block R2S or a block F2S depending on whether a direct rigid locking or a flexible joint is chosen (**Figure 1a**). We leveraged the advances in dual-material printing and fabricated a series of three building blocks through a commercial desktop-scale 3D printer (Ultimaker S3).^[43–46] Dual materials extruded are polylactic acid (PLA) as the base material for all blocks and thermoplastic polyurethane (TPU) as elastic joints for block F2S (**Figure 1b**), see **Figure S1** in Supporting Information. Once basic building blocks are fabricated, we then slot two S blocks into one block R2S such that a square unit cell S1 can be formed (**Figure 1c**). To connect various S1 units, a block F2S is required to replace one block S from each unit cell S1. For example, the design of a linear unit S3L is composed of two S blocks and three R2S blocks linked by two F2S blocks. Meanwhile, different base geometries (triangular, rectangular, hexagonal, etc.) can also be designed and assembled through the same modular approach. We illustrated a linear unit

using triangular building blocks in **Figure S3**, Supporting Information, but further discussion will be limited to square units for the sake of concept proofing.

Based on this assembly routine, a variety of planar units can also be formed from the single unit S1 to more complex 3D geometries. As a simple demonstration of our assembly strategy, we consider a baseline 2×2 square unit S4S composed of four R2S blocks and four F2S blocks as shown in **Figure 1d**. This assembly process is rather simple and straightforward from the block level to the unit level (**Figure S2**, **Movies S1** and **S2** in Supporting Information). By arranging these units horizontally, chain units can be designed and fabricated. As a representative example, the repeated chain unit S12-R has three S4S units aligned in the horizontal direction, which is essentially a linear design like unit S3L with an exception of expanding a single square unit to four subunits. We also designed and fabricated another chain unit S10-O by overlapping four S4S units with different predefined angles (90° , 0° , 90° , 0°). In **Figure 1e**, we further

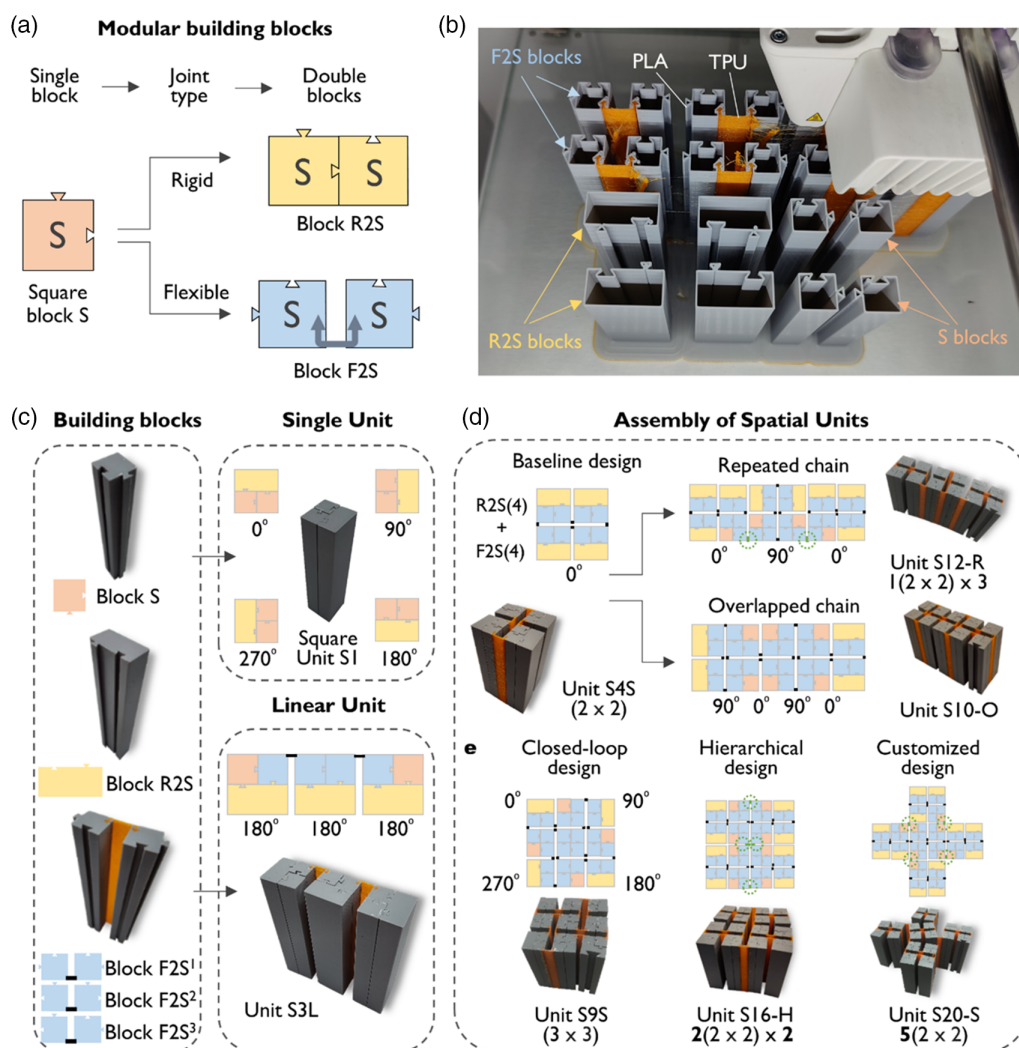


Figure 1. Programmable modular architected materials. a) Schematic and design logic of three building blocks. b) Snapshot of the dual-material printing process of building blocks. c) Fabricated building blocks and their assembly to form a basic single square unit (S1) and a 1×3 linear unit (S3L). d,e). Various planar units were generated from baseline unit S4S.

design three planar units by looping four S4S units (unit S9S), adding hierarchical features on S4S units (S16-H), and putting five S4S units in a spiral pattern (S20-S). These units will be discussed in the following sections for a specific function. More customized designs can be further exemplified by tailoring the allocation of these base units.

2.2. Conceptual Design of Adaptive Flow Regulators

Having demonstrated the versatility in the assembly of our modular architected materials, we next use them for a targeted application scenario related to flow regulation (Figure 2). Regardless of specific scales, traditional flow control usually relies on rigid structures. We started with simple linear units

with elastic joints and compared their behaviors against their rigid counterparts. In Figure 2a, we present four designs of flow regulators: a fixed baseline rigid unit (BR), a modular rigid unit (MR) built by five R2S blocks and two S blocks, and two modular flexible units (MF-S and MF-F) built by three R2S blocks, two F2S blocks, and two S blocks. Units MF-S and MF-F are distinguished by using either stiffer or softer joints of R2S blocks, respectively. Except for the BR unit, the other three modular units are connected with a bottom block through elastic joints. The bottom block is interlocked by a slot of a 3D-printed fixture, which is bolted to the base of the flow channel. To understand the deformation of these designs, we first conducted finite-element (FE) analyses using the commercial software ABAQUS 2020/Standard (Figure S4–S6 in Supporting Information).

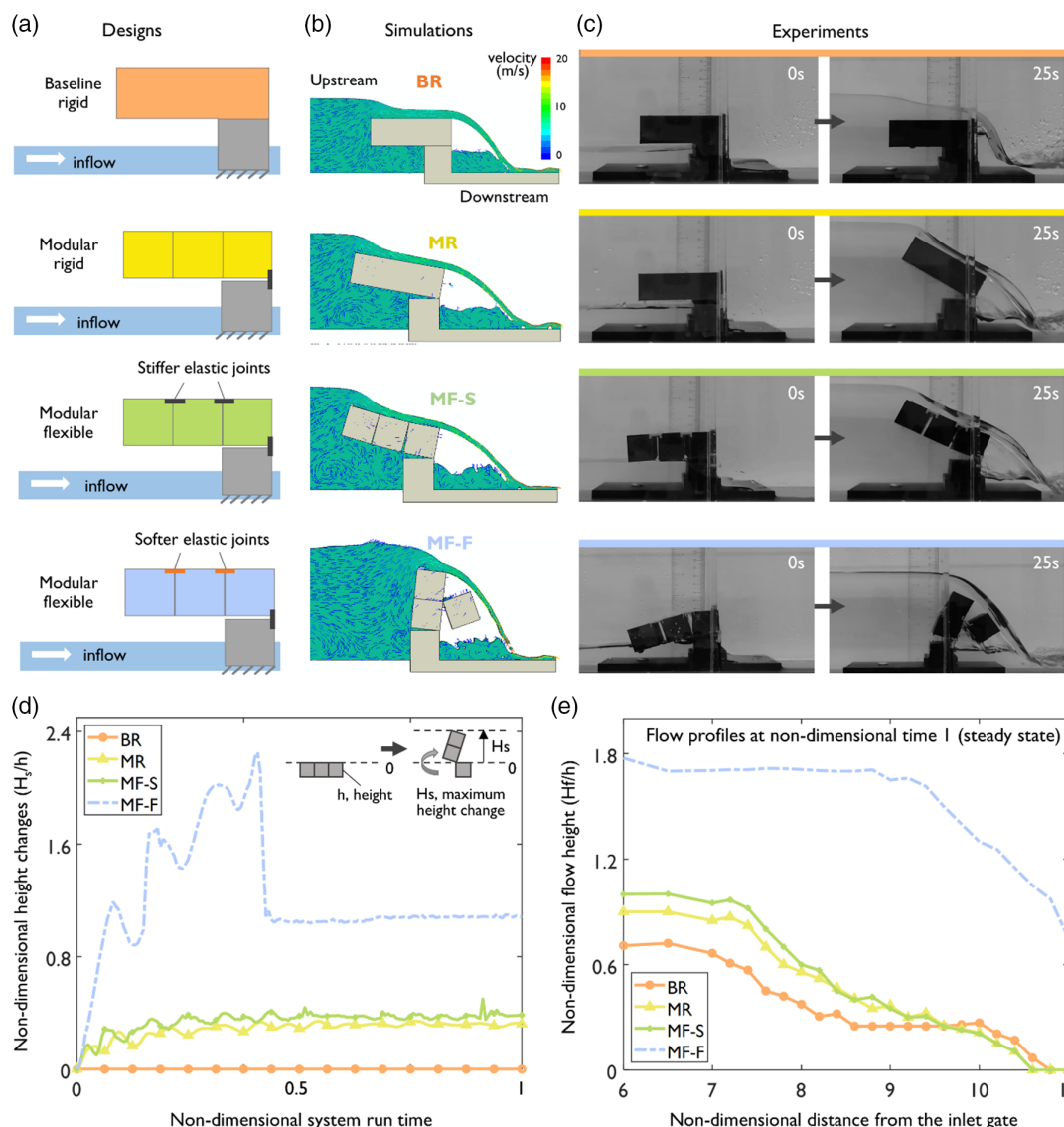


Figure 2. Design and characterization of flow regulators based on modular architected units. a) Schematic view showing four designs using various unit assembly. b) Snapshot of simulated behavior of regulators and velocity field. c) Initial and final state of regulators in experiments under the same flow rate. d) Simulated nondimensional height change ratio versus system time for four regulators. e) Simulated nondimensional flow surface profile for four regulators.

FE models were built by the Coupled Eulerian–Lagrangian method. The snapshots of four regulators after the overflow shown in Figure 2b (Movie S3, Supporting Information) indicate that with the inlet water coming from the left the behaviors of these flow regulators exhibited different features with the unit BR holding steady yet the other three units change their configurations to adapt to the pressure of the flow. As a result, the velocity vector field can be tuned along with the changes in the upstream volume of the regulator due to their shape-changing features.

To validate such reconfigurable features found in the simulations, we fabricated these units using a dual-material 3D printer, assembled them by hand, and tested their behavior in a small hydraulic flume (Figure S7 and S8 in Supporting Information). The observed behaviors in Figure 2c are in good agreement with simulated behaviors given in Figure 2b, providing experimental evidence to prove the validity of our designs and their tunability on the flow. To characterize the fluid–structure interactions, two height measurements (H_s and H_f) are conducted in simulations. H_s refers to the maximum height change measured from the upper surface of each unit during the reconfiguration. H_f refers to the absolute height of the stream measured from the bottom of the flume. We then plot a nondimensional run time versus dimensionless height changes (H_s/h) in Figure 2d to describe the behavior of the unit and a partial steady-state flow surface profile (H_f/h) to study the behavior of flow in Figure 2e for all four units. It can be seen that unit BR has no change in height while the other three units can transform due to the presence of elastic joints. Compared to the rectangular rigid unit MR, unit MF-S with two additional elastic joints has a slightly higher height change and steady-state flow surface profile, yet their overflow behaviors are quite similar. Reducing stiffnesses of joints allows unit MF-F to flip and entirely change the steady-state flow surface profile due to the shape transformation. With our simple modular approach, flow regulators with different characteristics can be realized by assembly and disassembly of a limited number of F2S blocks. Such advantage of the modular approach may not be obvious for just remaking another unit; in fact, a three-cell linear unit using the proposed modular approach would take a 57% longer time to be built and cost 50% more printing materials. However, our modular approach can save a considerable amount of production time and material used when a given unit becomes larger in terms of overall dimension and has a more complex geometry and joint layouts. For example, building a 12-cell unit can save at least 53% of the material consumption and 64% of the total printing time (see Figure S9 and Table S1 in Supporting Information for details).

2.3. Linear Units with Tailorable Properties

Based on the preliminary result provided earlier, in this section, we focus on three-cell linear units and investigate the effect of joint stiffness and geometry of unit cells on the fluid–structure interaction (Figure 3). Inspired by enriched geometric designs in polyomino, we considered our modular unit as the horizontal I shape in tromino (a polyomino of order 3) and easily designed five other units (a vertical I shape and four L shapes with different rotation angles). These shapes can have various transitions if

joints are appropriately assigned (Figure S10, Supporting Information). We first run numerical simulations on all six units with static geometries and found that the flow characteristics in terms of surface profiles (Figure 3a.) for those units can be categorized into four groups. Units I-0 and I-90 present as the lower bound and upper bound of surface profiles, while the other two pairs were bracketed in between and showed different flow characteristics.

To study the flow regulation induced by the reconfiguration of a single structure, we set the unit I-0 as an initial phase A and then tailored the stiffness of two horizontal elastic joints. The normalized stiffnesses (horizontal r_1 and r_2 ; vertical r_0) considered here range from 0 (a hinge joint) to 1 (a stiff joint). In Figure 2, we have identified two scenarios: when all joints are stiff ($r_0 = r_1 = r_2 = 1$), the unit MR-S stays in the same configuration and slightly rotates around the vertical joint as the center, while we observed two-step phase transitions (A to B to C) in unit MF-F when both horizontal joints become softer ($r_1 = r_2 = 0.02$). It should be noted that we used a stiff joint ($r_0 = 1$) for the vertical connection in the following scenarios, otherwise, units might easily flip over and other phase transitions would not occur. We thus started with two scenarios by setting one joint significantly stiffer than the other one ($r_1/r_2 = 15$ and $r_2/r_1 = 15$). Numerical results showed that under the condition of $r_2/r_1 = 15$, the left and the middle units will flip simultaneously (phase transition A to B), while only the left unit rotates if $r_1/r_2 = 15$, leading to a phase transition from A to C. Guided by the proven numerical model, we run a total of 81 simulations under different r_1 – r_2 combinations to build a phase diagram to distinguish all observed transition modes for the same linear unit under the same flow rate (Figure 3b, and S11 and Movie S4 in Supporting Information). Four transition modes discussed before have occupied four corners of the diagram. Since we are interested in two-step phase transitions (A to B to C) shown in the blue region, we tried to identify another two-step transition mode (i.e., A to C to B) but it is quite challenging since we found that the bending of joint one has a governing effect over joint two when the upstream flow comes from the left. In other words, the phase transition from A to B is easier to be triggered than the phase transition from A to C due to a larger bending moment applied to joint one than to joint two. Based on theoretical calculation (Figure S12, Supporting Information), we found a few r_1 – r_2 combinations that can trigger the targeted A to C to B transition and then form a purple region between the blue and green region.

To prove the accuracy of the phase diagram, we selected an r_1 – r_2 combination ($r_1 = 0.02$, $r_2 = 0.3$) from the orange region and verified the expected A–B phase transition numerically and experimentally in Figure 3c. To explore other factors controlling the shape transformation, we introduced an alternative joint arrangement by moving the joint two to the bottom of the unit and found that with a softer joint the unit can also achieve the same A–B phase transition. Although the geometry of the three-cell unit is quite simple, it has provided the effect of the joint stiffness on the deformation of chain design in which the phase diagram can be used for a more complex chain design, see Figure S13 in Supporting Information for a five-cell unit. To add more complexity to geometric features and potential tunability, we designed and fabricated two linear units with

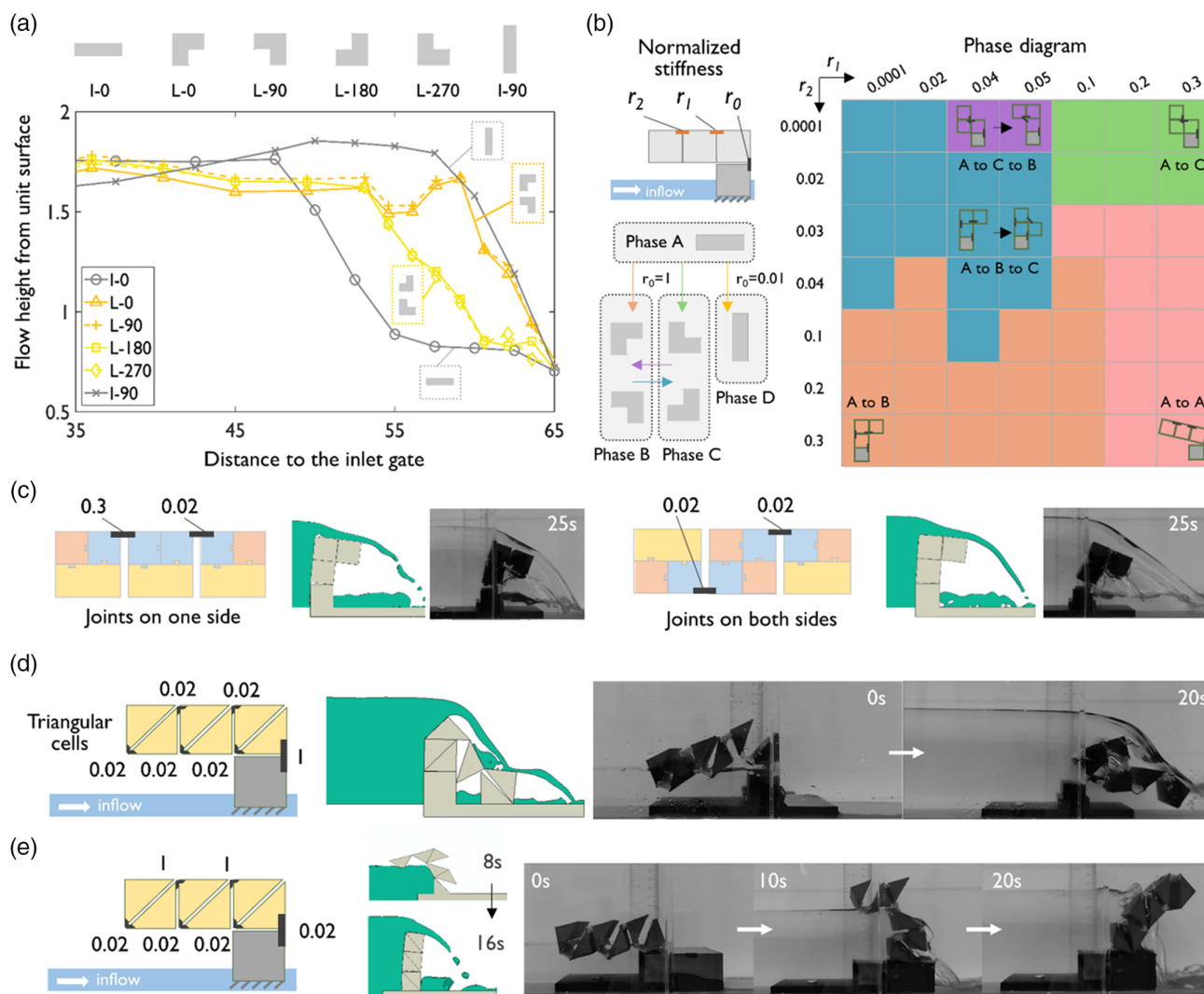


Figure 3. Linear units using modular architected materials achieve different flow regulations. a) Flow surface profiles by different regulators. Schematic view showing six designs with fixed geometries. b) Schematic joint design and resulting phase transition of the linear unit due to various combinations of joint stiffness. c) Effect of joint arrangements on the shape reconfiguration. d) Effect of cell geometries on the shape reconfiguration. e) Unit with programmable joints to achieve a horizontal to vertical shape-switching feature.

triangular tiles and compared their shape-changing features with the square unit counterpart. Figure 3d and Movie S5, Supporting Information, present the reconfiguration of a six-cell triangular unit with five soft horizontal joints and one stiff vertical joint, showing a similar phase transition from A to B yet adding more flexibility to the unit. Based on the proposed modular approach, we can easily replace joints with different stiffness and obtain a totally different function for the flow regulator. For example, in Figure 3e, two stiffer joints were used on the top while a softer one was used on the vertical joint. As a result, the new unit with the same number of triangular cells can flip smoothly in sequence and eventually form a vertical flow barrier, a favorable configuration to postpone the overflow. We did not systematically study the geometric design and joint arrangements for the triangular unit because we intend to focus on the modular design but we would expect a more interesting phase diagram with a new

geometry on the building blocks and their combinations. Nevertheless, our proposed modular approach shows robustness and versatility in controlling and tuning the flow for different purposes.

2.4. Planar Units with Programmable Configurations

To further illustrate the capability of our modular design, we introduced a series of closed-loop planar flow regulators with up to 10 units (Figure 4). In Figure 4a, three S4S units were designed to identify the effect of joint stiffness on the rotation of such planar units. Specifically, unit S4S-A has four hinge joints ($r = 0.0001$); unit S4S-B has four elastic joints ($r = 0.02$); and unit S4S-C has three hinge joints plus one stiffer joint ($r = 0.08$). Numerical results suggested that the same unit can undergo a varied angle of rotation with different joint

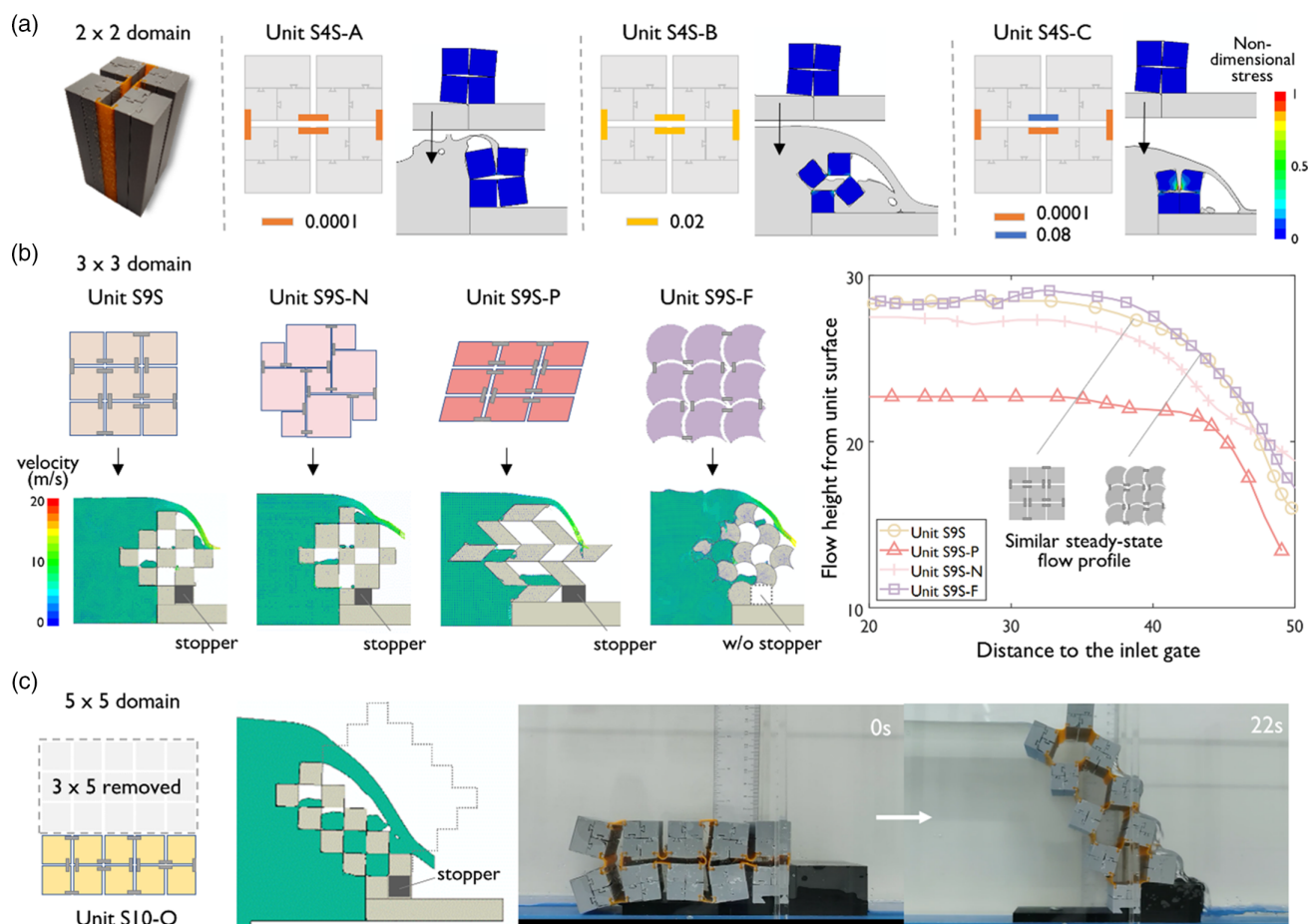


Figure 4. Planar units using modular architected materials. a) Schematic view showing three four-cell designs with varied joint stiffnesses and their predicted behaviors under the same flow rate. b) Schematic view showing four nine-cell designs with varied cell geometries, their predicted behaviors, and corresponding flow surface profiles. c) A 10-cell chain unit showing a vertical expansion feature to postpone the overflow.

combinations but the unit S4S-B exhibited a more gradual behavior without a noticeable stress concentration shown in the stiff joint of the unit S4S-C (Figure S12, Supporting Information). Harnessing these elastic joints, we then designed four closed-loop 3 by 3 units with four types of cell geometries (square, nonuniform square, rhombus, and fan) and compared their reconfigurable features using FE simulations (Figure 4b and Movie S6, Supporting Information). To maintain the maximum height of each unit, we added a stopper adjacent to three units except for the fan design so that further rotations can be restrained. Our numerical results demonstrated their transformed shapes along with the flow surface profiles under the same run time. The flow behavior and overflow velocity are also varied mainly because of different geometric contours after the expansion of these units. As expected, all units fully expanded and rested for three units after the contact with the stopper while unit S9S-F with fan-shaped cells exhibited a self-locking phenomenon due to its geometric feature and achieve a similar steady-state flow surface profile like the unit S9S with a stopper. To explore a wider expansion of the regulator, we designed a 25-cell closed-loop planar unit (Figure 4c) and numerically proved its flow tuning behavior but the unit cannot be

installed in the hydraulic flume. We thus removed 15 cells on the top and obtained a chain-like planar unit S10-O. Interestingly, this unit can transform into an inclined shape that is three times higher than its original height. If more cells are added horizontally, the height of the unit and the corresponding flow characteristics can be further tuned for a targeted behavior in a specific application scale.

Having confirmed the validity of our linear and planar units, we seek to combine them for more advanced assembly (Figure 5). Using the four-cell unit S4S with hinge joints as the base geometry, we embedded them into linear and planar patterns through a hierarchical design approach as shown by two representative units in Figure 5a. In Figure 5b,c, we presented two 1×3 linear units using three S4S units connected by two soft elastic joints. In our simulation (Figure 5b and S13, Supporting Information), unit S12-R1 exhibited a two-step phase transition that was observed in the blue region of the phase diagram (Figure 3b and Movie S7, Supporting Information) for the previously mentioned three-cell linear unit S3L, yet the hierarchical design enables local shape-changing that leads to extra unevenness on the contour of the flow regulator. In Figure 5c,d, we successfully designed units S12-R2 and S16-R to achieve a targeted inclined

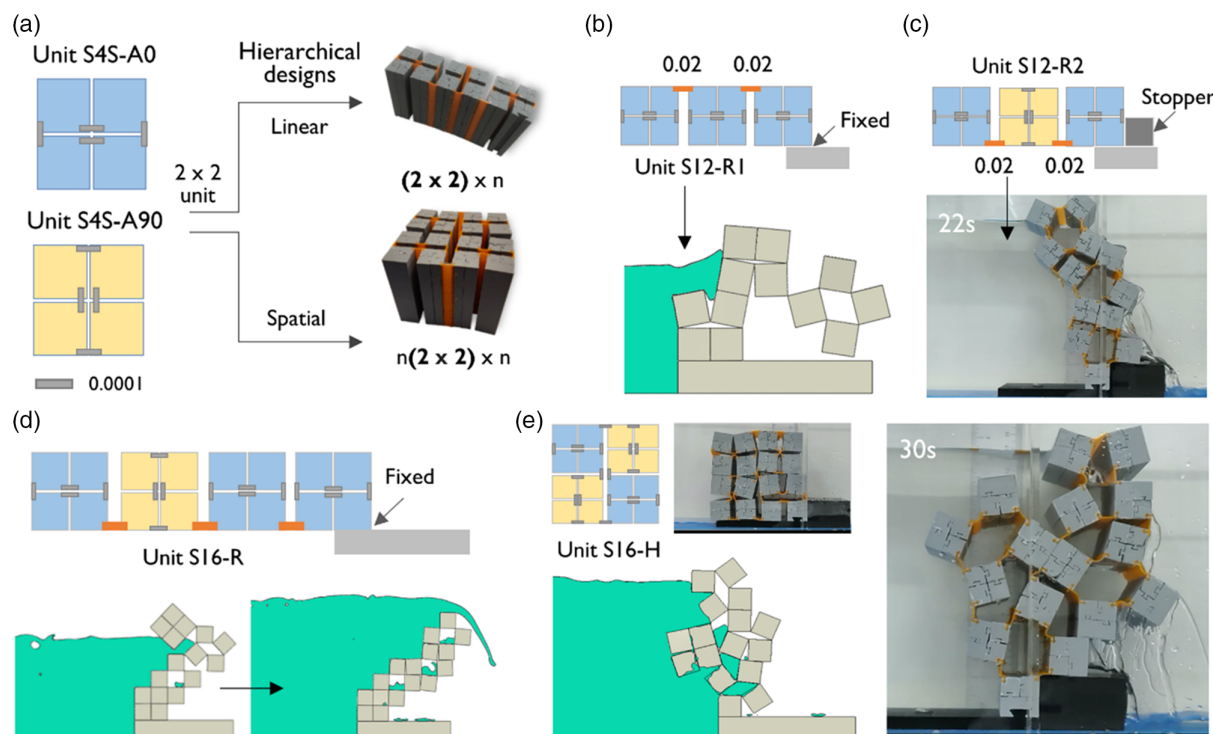


Figure 5. Hierarchical designs of modular architected materials. a) Schematic view showing a linear and a planar unit assembled from baseline four-cell units. b) Design and simulation of a 12-cell linear unit under the flow pressure. c) Experimental snapshot showing the inclined expansion of a 12-cell unit. d) Design and phase transitions of a 16-cell linear unit. e) Numerical and an experimental snapshot of a 16-cell planar unit showing expansion.

sequential expansion of each S4S unit. Similar to unit S10-O, if more cells are added to the left side of the unit, it can further grow to a higher regulator to postpone the overflow as long as the joint at the bottom of the unit can sustain a larger bending moment. Enabled by our assembly strategy, unit S16-R can be reassembled to a planar unit S16-H for regulating overflow other than avoiding them. As a final demonstration, we confirmed such lateral expansion of the unit and our numerical and experimental results showed a good agreement in terms of shape-changing characteristics.

3. Conclusion

In summary, we have presented a new class of modular architected materials based on a simple easy-to-assemble strategy and proved its versatility and robustness in regulating flow. Our numerical and experimental results showed that by programming cell geometry, unit distributions, and the spatial arrangement of joints, a series of linear and planar flow regulators were designed and built. The architected units can be assembled in a variety of forms with tailorable stiffness and controllable reconfigurations, leading to a multifunctional flow regulator that can tune the upstream flow, postpone the overflow, control the flow velocity and flow surface profile, etc. We envision the proposed integrated material-structural design concept can be expanded to realize adaptive flow regulations over a wide range of scales, from microfluidic devices to large-scale weirs or flood barriers.^[47] Advanced programmability and self-assembly can be

further enabled by coupling with a variety of intelligent materials and multi-material printing techniques.^[48–50] Another route lies in the inverse design of architected materials using artificial intelligence and machine learning for specific fluid flow situations and performance targets.^[51–53] Overall, we demonstrated a new route of assembling modular architected materials for flow regulation but our strategy is also applicable for instability-based, auxetic, kirigami-based, and other types of architected materials that might be applied to a broader wave control applications such as acoustics.^[54–56]

4. Experimental Section

3D Printing: Single- and multi-material architected units were printed using the fused deposition modeling method with a nozzle diameter of 0.4 mm (Ultimaker, S3 Utrecht, Netherlands). Dual materials chosen were PLA as rigid blocks and thermoplastic polyurethanes (TPU) as elastic joints. To prevent extruding problems in dual extrusion, a Prime Tower was printed with each batch of units. The height of the printing layer was set at 0.2 mm, while the wall thickness was set at 0.8 mm. For the print speed, we selected 40 mm s^{−1} for PLA and 20 mm s^{−1} for TPU. For the nozzle temperature, we set 205 °C for PLA and 225 °C for TPU. An infill density of 0% was chosen to accelerate the production time and reduce the self-weight of units for an easier rotation. Width of basic building blocks was 2.5 mm (S), 5 mm (R2S), and 5 mm plus the joint length (F2S). All assembled linear units discussed in Figure 2 were 99.7 mm long, 0.3 mm shorter than the width of the channel to avoid large friction force on the sidewall. The dimension of units with triangular cells was 75 × 25 × 99.7 mm. In Figure 3, the four-cell square unit S4S was 58 mm long while the unit S10-O had a total length of 157 and a height

of 58 mm. Lastly, as shown in Figure 5, the side of the unit S16-H was 124 mm long.

FE Analysis: The commercial FE software Abaqus was used to numerically investigate the response of the architected material units. A Coupled Eulerian–Lagrangian method was applied to consider the fluid–structure interaction, see Supporting Information for detailed model parameters and descriptions. In brief, the Eulerian domain represents the region that the liquid can flow while the Lagrangian domain simulates the behavior of structures. The Abaqus/Explicit solver was employed for all the simulations.

Hydraulic Flume Tests: Experimental tests were conducted in a self-circulate open channel flume with a width of 100 mm and could provide a constant flow discharge of 5 L s^{-1} . The bottom block of each specimen had a notch that could be slotted into a 3D-printed fixed part that was attached to the channel.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

adaptive structures, architected materials, dual printing, flow regulator, origami

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